

Transformer MLP Gate Thresholds Are Couplings to a Carried Reference Direction

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Abstract

The corpus-mean direction of a transformer’s residual stream is usually treated as a nuisance and subtracted before analysis. We present evidence that it is a functional component: the reference against which the MLP gate population sets its operating point. In Phi-2, an exact decomposition of resting gate pre-activations shows that the resting inhibition of $>99.9\%$ of gates is carried by the coupling $w \cdot b_L$ to the carried mean direction, at $50\text{--}60\times$ the explicit bias parameter, and this resting term rank-predicts each neuron’s firing rate at $\rho \approx 0.95$. Causally, removing the stream’s projection on the reference multiplies above-threshold firing by $\sim 9\times$, dose-monotonically, at $30\text{--}40\times$ a norm-matched control; a random-initialised twin is flat, and replacement tests show the function is carried by the direction, not the magnitude or token content. The decomposition replicates on four further families spanning both gate types (GELU with an explicit gate bias, bias-free SwiGLU), and the dose-response on two of them. Across an eight-model scan the mechanism is present in every GELU and SiLU family and absent only in OPT, where an opposing LayerNorm bias cancels the carried reference. Gate thresholds are implemented as couplings to a constant the network itself builds and maintains, not as bias parameters.

1 Introduction

Transformer residual streams contain large, stable components that are shared across inputs: attention sinks and massive activations (Xiao et al., 2023; Sun et al., 2024), and outlier dimensions tied to token frequency (Kovaleva et al., 2021; Timkey and van Schijndel, 2021; Puccetti et al., 2022). Existing work locates these objects and shows they matter for quantisation and attention. What they are *for*, computationally, is less settled.

This paper studies one such object, the corpus-mean direction of the layer-normalised residual stream, b_L , and measures a specific function for it: it is the reference against which the MLP gate population sets its operating point. The claim decomposes into parts that were tested separately, each against a random-initialised control model (“twin”) of identical architecture:

1. **Object** (§2): b_L is constant across text, is nearly a single direction shared across mid-stack layers, and its mid-stack content is not the surface function-word distribution.
2. **Function** (§3): removing or amplifying the stream’s projection on b_L releases or suppresses gate firing, dose-monotonically, far beyond norm-matched controls; the twin is flat.
3. **Identity** (§4): the function is carried by the direction, not by magnitude or token content.

4. **Implementation** (§5): resting gate inhibition is the coupling $w \cdot b_L$, not the bias parameter. This is an exact decomposition, not a fit.
5. **Maintenance and dependence** (§6): the direction is rebuilt by aligned per-layer mean writes after local removal; defeating restoration by co-removal is near-catastrophic for prediction.
6. **Formation** (§8): on public Pythia checkpoints, the reference coupling appears in the same early window as gate sparsification and induction heads.

The battery is characterised in Phi-2, with Formation on Pythia-410M checkpoints. The two most portable legs, the static decomposition (§5) and the causal dose-response (§3), are then tested on further families; the remaining causal tests are Phi-2-only (Limitations).

Throughout, “gate” means the scalar nonlinearity of an MLP hidden unit; “duty cycle” is the fraction of positions at which a unit’s activation is positive. Two background facts motivate the hypothesis. First, the trained model’s median duty cycle is 0.076 against 0.48 at random init: trained gates are off by default (Szatkowski et al., 2025), and “off by default” is only defined relative to what occupies the stream at rest. Second, in a decomposition of what drives firing, the corpus-mean component contributes a share of about -0.40 of trained drive at every layer measured, and ≈ 0 in the twin: trained gates are held down by something constant. The experiments below ask whether that constant is functional structure.

Provenance. Every experiment was pre-registered with named outcome branches and construction-validity gates (each instrument must first reproduce a known result before its new reading is taken); one pre-registered prediction failed and is reported as failed (§7). Primary model: Phi-2 (2.7B, 32 layers, $d=2560$), fp16, hooks on public weights; twin: same architecture, random init. Cross-model replication: GPT-2-medium, Pythia-1.4B, Qwen2.5-1.5B, TinyLlama-1.1B, plus an eight-model GELU/ReLU/SiLU scan, each against its own `from_config` twin. Formation: Pythia-410M-deduped public checkpoints (Biderman et al., 2023).

2 The object

Definitions. m_L is the corpus-mean raw residual at the input of layer L ; b_L is the corpus-mean layer-normalised MLP input at L (the object gates actually read); $\hat{m}_L$ the unit vector of m_L . Statistics are computed over document positions ≥ 16 , excluding the attention-sink region.

Constancy. Split-half cosine of b_L across document halves is 0.94–0.99 at every measured layer.

One direction, not one per layer. $\cos(\hat{m}_{10}, \hat{m}_{18}) = 0.94$: the mid-stack reference is close to a single shared direction. Functionally this is confirmed in §4: layer 10’s gates accept layer 18’s mean direction as their reference almost without loss. This cross-layer alignment is a Phi-2 property, not a universal one (it is family-dependent across the replication models; Limitations); every other claim in the paper is per-layer and stands without it.

Content. Decoded through the unembedding, mid-stack b_L (L2–L18) is only weakly token-shaped (max row cosine 0.14–0.15, about $1.7\times$ the isotropic floor), and its nearest tokens are not function words but truncated word stems (*dilig*, *conclud*, *showc*, *withd*, ...). The high-frequency function-word surface (*the*, *and*, *a*, *to*, ...) appears in b_L only at L22–L26, close to the readout. The natural prior, that the constant

α	$\Delta\text{frac}(\text{act}>0.5)$			KL vs. clean		
	L6	L10	L14	L6	L10	L14
0.0 (removed)	+0.103	+0.086	+0.075	1.427	0.519	0.274
0.5	+0.023	+0.023	+0.021	0.047	0.040	0.035
1.0 (identity)	0.000	0.000	0.000	0.000	0.000	0.000
1.5	−0.006	−0.007	−0.007	0.014	0.018	0.020
2.0 (doubled)	−0.008	−0.009	−0.009	0.055	0.060	0.110
norm-matched control	+0.003	+0.001	+0.003	0.020	0.022	0.025

Table 1: Dose-response of gate firing and output damage to scaling the stream’s projection on $\hat{m}_L$ at the manipulated layer (Phi-2, trained). The twin is flat at every α ($|\Delta\text{frac}| \leq 0.002$, no ordering). Point estimates are document medians over 39 documents; 95% document-bootstrap CIs (2000 re-samples) on the removal row: Δfrac $[+0.096, +0.112]/[+0.079, +0.098]/[+0.068, +0.088]$ and KL $[1.14, 2.03]/[0.36, 0.75]/[0.15, 0.40]$ (L6/L10/L14), each disjoint from the norm-matched control’s KL CI ($\leq [0.010, 0.032]$). The document-sampling spread on KL dwarfs the run-to-run fp16 jitter ($\sim 1\%$), so only the leading two digits are informative.

component of English text’s representation is the function-word bed, is wrong mid-stack; the reference is a largely off-vocabulary direction.

Sources. The embedding mean is negligible (norm 0.42, cosine 0.02 to the deep mean). Instead, every layer’s attention *and* MLP mean writes are mutually aligned with the deep mean (cosines $+0.2$ to $+0.7$, never negative), each small (norms ~ 1 – 3 against update RMS 15 – 38), summing coherently: the reference is a distributed construction to which every component contributes a small aligned DC term.

3 Function: the gate population thresholds against it

Manipulation. At layer $L \in \{6, 10, 14\}$, the residual entering the layer is modified at all positions ≥ 16 : $x' = x + (\alpha - 1)(x \cdot \hat{m}_L) \hat{m}_L$, for $\alpha \in \{0, 0.5, 1, 1.5, 2\}$. $\alpha=1$ is the identity (verified as an exact no-op, KL = 0); $\alpha=0$ removes the stream’s projection on the reference; $\alpha=2$ doubles it. The control subtracts the same per-position magnitude along a fixed random unit direction (norm-matched). 39 documents.

Result. Table 1 shows the dose-response. Removal releases firing; amplification suppresses it; the response is monotone through the identity point at every layer and 30 – $40\times$ the norm-matched control, consistent with a population of near-binary gates driven across threshold together (Balogh, 2026). Against the clean baseline of $\text{frac}(\text{act}>0.5) = 0.0105$ at L10, the release is a $\sim 9\times$ multiplication of above-threshold firing. The suppression side saturates small, which is the expected asymmetry: at duty 0.076 the population is already near its inhibited floor, so added inhibition has little left to turn off while release frees every marginal gate. The twin shows no response, having no resting coupling to lose (§5).

Output damage. Removal at a single site costs KL = $1.43/0.52/0.27$ (L6/L10/L14) against 0.020 – 0.025 for the norm-matched control. The twin loses only 0.32 – 0.38 KL under the same removal despite its raw mean being 8 – $11\times$ larger in norm (168 vs. 15 – 21): per unit of removed norm the trained model is roughly two orders of magnitude more sensitive.

replacement $\hat{c}$	$\cos(\hat{c}, \hat{m}_{10})$	$\Delta\text{frac}(\text{act}>0.5)$	KL
none (pure removal)	—	+0.0845	0.521
coordinate-shuffled $\hat{m}$	−0.003	+0.0840	0.963
random direction	+0.011	+0.0778	0.502
function-word W_U sum	+0.096	+0.0740	0.402
layer 18’s $\hat{m}$	+0.940	+0.0027	0.004
identity	1.000	0.000	0.000

Table 2: Replacement panel at L10 (Phi-2, 23 documents). Preservation of quiet firing tracks $\cos(\hat{c}, \hat{m})$ monotonically.

Restoration. The release is local: downstream firing is undisturbed. After removal at L6, $\cos(\text{resid}, \hat{m})$ is down -0.16 at L10 and L14, -0.10 at L18, and recovers to baseline by L26–L30: the stream re-accumulates its mean over ~ 12 – 16 layers, consistent with the anatomy (every layer’s mean write is reference-aligned, so the reference self-heals by accumulation). Deeper removal sites restore more slowly.

Cross-model. The same knob at one mid-stack layer (depth ≈ 0.31 : Pythia-1.4B L8, Qwen2.5-1.5B L9; 29 documents, each against its own twin) reproduces the sign, the monotonicity, and the control separation. On the duty-cycle readout (activation > 0), removal shifts duty by $+0.318$ $[0.303, 0.329]$ (Pythia) and $+0.302$ $[0.286, 0.316]$ (Qwen), doubling by -0.124 and -0.081 , monotone across a 15-point α grid, against $+0.0001$ for the norm-matched control and flat twins ($|\Delta\text{duty}| \leq 0.002$); Phi-2 on the same measure is $+0.238 / -0.070$. Qwen is the sharpest case: its SwiGLU gate has no bias parameter, so the carried reference is the entire resting inhibition, and scaling it still moves firing in dose-response. Removal costs final-position KL 3.60 (Pythia) and 0.59 (Qwen).

4 Identity: the direction, not the magnitude or the tokens

Removal tests necessity of the projection, but it leaves two alternative readings open. Gates might not read this direction at all and simply need the stream to be full; and since the manipulation acts on the pre-norm residual, removing the projection shrinks the norm entering LayerNorm, so the renormalisation boosts the gain on every orthogonal component, so some of the release could be that gain boost rather than the loss of the inhibitory coupling. The replacement panel addresses both: at L10, $x' = x - (x \cdot \hat{m})\hat{m} + (x \cdot \hat{m})\hat{c}$ transfers the signed per-position projection onto a candidate direction $\hat{c}$; magnitude, sign pattern, and timing are preserved, the total norm is approximately unchanged (the cross-terms with $\hat{c}$ are small), and only the direction changes.

In Table 2, shuffled and random directions release firing exactly like removal; a unit sum of function-word unembedding rows (cosine 0.096 to $\hat{m}$) buys a sliver proportionate to its cosine and nothing more, so token content per se contributes nothing; and another layer’s mean direction (cosine 0.94) preserves quiet firing almost perfectly, $31\times$ less release and $130\times$ less KL than removal. The magnitude-reader alternative is excluded: no low-cosine candidate preserves anything. The gain-boost alternative is excluded the same way: a random-direction replacement leaves the pre-norm magnitude in place, so the LayerNorm gain is the same as clean, yet firing releases as if the projection had been removed, while layer 18’s direction, under the same norm, preserves quiet firing. One further observation: the shuffled replacement’s KL (0.96) *exceeds* removal’s (0.52); transferring the projection onto a wrong direction injects a coherent error signal on top of losing the reference.

layer	$w \cdot b_L$ (median)	b_n (median)	resting	ref-dominant	$\rho(\text{resting, duty})$
6	-1.535	-0.027	-1.568	99.98%	0.943
10	-1.226	-0.022	-1.250	99.99%	0.959
14	-1.091	-0.021	-1.115	99.96%	0.960
22	-1.130	-0.018	-1.149	99.94%	0.954

Table 3: Exact resting decomposition of gate pre-activations (Phi-2, 19 documents). “Ref-dominant” is the fraction of gates whose resting inhibition is carried mostly by $w \cdot b_L$. 95% document-bootstrap CIs (2000 resamples): ref-dominant stays $\geq 99.89\%$ at every layer (widest $[99.89, 99.98]$ at L22), and ρ is $[0.936, 0.945]/[0.951, 0.959]/[0.953, 0.959]/[0.945, 0.952]$ (L6/L10/L14/L22).

5 Implementation: the threshold is the coupling

A gate’s *resting* pre-activation decomposes exactly: $\langle w \cdot x_{\text{in}} + b_n \rangle = w \cdot b_L + b_n$, since $\langle x_{\text{in}} \rangle = b_L$ by definition. If gates threshold against the carried reference, training should have moved the operating-point inhibition into the coupling $w \cdot b_L$ rather than the explicit bias parameter b_n .

Table 3: the reference coupling carries the resting inhibition at 50–60 $\times$ the explicit bias parameter for more than 99.9% of gates at every measured layer, and the resting pre-activation rank-predicts each neuron’s duty cycle at $\rho \approx 0.95$. The gate population’s sparsity *is* the reference coupling, almost completely. In the twin, $w \cdot b_L \approx 0$ (median magnitude ≤ 0.02) and duty is 0.5; its bias parameters are exactly zero at init, so the informative twin statistic is the absent coupling. This is the classical bias-as-weight-to-a-constant-input construction, discovered in the weights: the model implements thresholds as couplings to a constant it itself carries and maintains. It also explains the dose-response of §3 without further assumptions: scaling the stream’s projection on $\hat{m}$ scales the operating-point term of every gate simultaneously.

Per-gate anatomy. Factoring each fc1 row against the unit reference, $w = c_n \hat{b}_L + w_\perp$, the coupling is a small slice of read norm: $|c_n|/\|w\|$ has median 0.07–0.12 across layers against a random-twin floor of 0.013, thinning toward the readout, so ≈ 0.99 of the row remains free for content. The coupling alone rank-predicts duty as well as the full resting inhibition ($\rho(c_n, \text{duty}) = 0.943\text{--}0.959$, matching Table 3); b_n adds nothing to the ordering. Per gate, the coupling is a low single-digit-percent tax on the read weight that fixes where the gate sits while leaving its content direction almost untouched.

The coupling over training. The decomposition above compares two endpoints. Pythia-410M-deduped, whose GELU gates carry an explicit bias and whose checkpoints are public, lets the two static terms be tracked step by step (four mid-stack layers). b_n is exactly zero at initialisation (Pythia zero-inits biases) and stays ≤ 0.04 in magnitude the entire run, while $|w \cdot b_L|$, after training first destroys the init constant ($\sim 0.25 \rightarrow 0.12$ by step 128), builds through the step 512 $\rightarrow$ 1000 window ($0.16 \rightarrow 0.38$), peaks near 0.85–0.95 around step 4000, and relaxes partially thereafter. The reference coupling carries the resting inhibition of $\geq 99.6\%$ of gates at every checkpoint including init: there is no phase in which the explicit bias leads and is overtaken. The network never uses the bias parameter at all; it builds the coupling directly. Duty falls $0.51 \rightarrow 0.16$ over the same build and $\rho(\text{resting, duty})$ stays 0.91–0.998 throughout. The endpoint inference is an observed time-course, and it independently reproduces the formation window of §8 through a different statistic.

model	gate (bias?)	resting med.	ref-dom.	$\rho(\text{rest, duty})$
GPT-2-medium	c_fc (yes)	−1.52 to −0.99	99.4–99.7%	0.91–0.97
Pythia-1.4B	h_to_4h (yes)	−1.08 to −0.74	> 99.9%	0.96–0.98
Qwen2.5-1.5B	gate_proj (no)	−4.14 to −0.37	—	0.62–0.99
TinyLlama-1.1B	gate_proj (no)	−0.21 to −0.07	—	0.97–0.99

Table 4: Cross-model resting decomposition (30 documents, four mid-stack layers per model at depth fractions 0.2–0.65; ranges span those layers). “Ref-dom.” is the fraction of gates whose resting inhibition is carried mostly by $w \cdot b_L$ rather than the bias parameter (vacuous for the bias-free SwiGLU gates; for these the informative statistic is the negative resting coupling and its duty-prediction). Where a bias exists, $|w \cdot b_L|/|b_n|$ is 15–18 $\times$ (GPT-2) and 38–43 $\times$ (Pythia). Every model’s random-init twin is floored: resting median $|\leq 0.01|$, duty ≈ 0.5 .

Cross-model replication. The resting decomposition is static (weights and mean activations only), so it can be re-run cheaply on other families against a `from_config` twin. Table 4 does so on four more architectures, spanning both gate types: GELU MLPs with an explicit gate bias (GPT-2-medium, Pythia-1.4B) and bias-free SwiGLU gates (Qwen2.5-1.5B, TinyLlama-1.1B). In every family the resting pre-activation is negative (gates held off by the carried reference) and rank-predicts duty at $\rho \geq 0.96$ across mid-stack layers, with the twin floored (resting ≈ 0 , duty ≈ 0.5) throughout. Where a gate bias exists, the reference coupling dominates it for > 99% of gates at 15–43 $\times$ the bias parameter; where the gate is bias-free, the coupling alone carries the entire negative resting term. The one sub-threshold ρ is Qwen’s shallowest sampled layer (L_5 , $\rho = 0.62$, a strongly suppressed near-off regime, duty 0.004); every deeper layer is $\rho \geq 0.96$. Magnitude varies (TinyLlama’s resting inhibition is an order of magnitude weaker than GPT-2’s).

A counteracting boundary case. The mechanism is not universal. Extending the static decomposition to eight further models spanning GELU, ReLU, and SiLU gates, it is present in all six non-OPT families (ref-dominant 99.5–100% where a bias exists, $\rho \geq 0.89$ throughout) but absent in both OPT models (opt-350m, opt-1.3b): their resting inhibition is 40–130 $\times$ weaker, bias-dominated, and rank-predicts nothing ($\rho = 0.000$). A norm decomposition explains why. Writing $w \cdot b_L = w \cdot (b_L - \beta) + w \cdot \beta$ (residual-maintained mean plus LayerNorm bias), the residual-maintained part is primary wherever the mechanism is present: it dominates 91–99% of gates in the GELU LayerNorm models and equals the resting term by construction in the bias-free RMSNorm models, with β a 7–26% augmentation. So the §6 attribution of the reference to a maintained residual mean, rather than a static norm parameter, holds. OPT is the exception that locates the requirement: it builds a large residual-carried inhibition comparable to GPT-2’s but trains an almost exactly opposing LN-bias excitation ($|w \cdot (b_L - \beta)| \approx |w \cdot \beta| \approx 0.5$ –0.6, opposite sign), netting $w \cdot b_L \approx 0.006$: a reference cancelled at the operating point, not absent. ReLU-versus-family stays formally confounded (every ReLU model sampled is an OPT).

The read side. A separate census of Phi-2’s conjunctive gate neurons finds that most carry an inhibitory reader: a token whose presence drives the gate off. Those suppressor-token directions align with $\hat{b}_L$ at a mean-removed cosine of 0.28–0.52 across eight layers, while content-token and isotropic directions are orthogonal to it (aggregate −0.001 and 0.003), and the alignment vanishes against the twin’s reference (≤ 0.02). Read-side “gate off when a suppressor token is present” and write-side resting inhibition ($w \cdot b_L < 0$) are one fact: the gate population couples negatively to the carried reference.

6 Maintenance, and the cost of losing the reference

Attention carries the steady feed. Decomposing each block’s output into its component along $\hat{m}$ and the orthogonal remainder, and measuring the constant (split-half) share of each: at L10, the attention block’s $\hat{m}$ -channel has constant share 0.452 against 0.014 for its orthogonal channel, a $33\times$ dissociation; the block outputs a steady reference feed while its content channel fluctuates. The MLP shows no dissociation (0.003 vs. 0.003), consistent with the earlier finding that per-layer MLP DC components mutually cancel. In the twin both channels are largely constant (0.93 vs. 0.87; an untrained block is a mostly-constant filter), so the dissociation is a trained property. Together with §2: per-layer mean writes build the reference; attention sustains it in operation.

Co-removal. Single-site removal understates the dependence because the stream heals. Removing the projection at *all* measured layers ≥ 6 simultaneously releases firing at every depth (at L18 more than at the shallowest site) and costs $\text{KL} = 4.38$, $\Delta\text{NLL} = +4.75$ nats, about $8.5\times$ the single-site effect, while the twin under the identical intervention loses 0.65 KL with flat firing (document-bootstrap 95% CIs over 23 documents: $\text{KL} [3.74, 6.05]$, $\Delta\text{NLL} [4.62, 5.03]$, disjoint from the twin). Late-only co-removal (≥ 14) is far milder ($\text{KL} 0.55$, CI $[0.24, 0.90]$): the dependence is concentrated mid-stack. The reference is a standing operating condition of the trained trunk, not a local convenience of any layer.

7 Operating point

A pre-registered prediction failed here: we expected the reference to act as a signed excitability knob, with output entropy rising monotonically in the reference magnitude. It does not. On a finer α grid at L10 ($\alpha \in \{0.6, \dots, 1.4\}$, 29 documents, document-bootstrap CIs, position terciles), entropy has its minimum exactly at the natural magnitude ($\Delta H = 0$ at $\alpha=1.0$, $\Delta H \geq 0$ elsewhere, monotone on each arm), in every position tercile, twin floored. The two arms are unequal. The release arm ($\alpha < 1$) clears the norm-matched control with disjoint CIs and is universal: it replicates on Pythia-1.4B and Qwen2.5-1.5B (ΔH at $\alpha=0.8$: +0.057 Pythia, +0.047 Qwen, +0.013 Phi-2, each above a control $\leq |0.0005|$). The amplification arm is family-dependent: symmetric in Pythia, shallow in Phi-2 (control-separated only for $|\alpha - 1| \geq 0.3$), and mildly entropy-reducing in Qwen over $\alpha \in [1.05, 1.3]$. The robust reading is a floor from below: the natural reference magnitude is the confident point, and weakening the reference always floods entropy in.

8 Formation

On Pythia-410M-deduped public checkpoints (different family and scale from the causal experiments; one seed), three statistics move together in the step 512→1000 window: the induction-head score (0.03 → 0.88), the median duty cycle (0.42 → 0.28, continuing to 0.16 by step 4000), and the reference share of resting drive ($-0.06 \rightarrow -0.34$, deepening to ≈ -0.75 by step 3000–4000). The reference frame and gate sparsification form as one construction, in the same window as the induction transition (Olsson et al., 2022). The joint timing of the duty-cycle collapse (activation sparsity is a known endpoint property (Szatkowski et al., 2025)) with induction and readout reorganisation is developed in a companion paper (Tuomi, 2026); the reference-formation reading is added here. One further observation: training’s first measurable act on the MLP write is destroying the init constant (constant share 0.40 → 0.15 by step 128, before anything else moves), so the trained reference is a construction, not a residue of initialisation, consistent with the training dynamics of massive activations (Gallego-Feliciano et al., 2025). The claim

“jointly” is bounded by checkpoint resolution: 512→1000 is one interval, and the induction transition itself unfolds in multiple phases (Minegishi et al., 2025; Xu, 2026); checkpoints below step 512 exist and are the direct test, not yet run.

9 Related work

Massive activations, sinks, outlier dimensions. Attention sinks (Xiao et al., 2023) and massive activations (Sun et al., 2024) are large constant components with a bias-like role in attention; Sun et al. (2024) explicitly propose them as bias terms (see also Owen et al., 2025), and later work ties sinks to compression valleys (Queipo-de Llano et al., 2025), to outlier-driven rescaling essential for training (Qiu et al., 2026), and to depth-transient damping (Hengyu, 2026). Outlier dimensions are frequency-driven and normalisation-amplified (Kovaleva et al., 2021; Timkey and van Schijndel, 2021; Puccetti et al., 2022), and Macocco et al. (2025) show that last-layer outlier dimensions act as a constant positive logit contribution toward frequent tokens. The object studied here is in the same family but the measured function differs in site and form: a *direction* (not a coordinate set), read at the MLP gate inputs mid-stack (not at the output logits), that carries the resting inhibition of the gate population by exact decomposition, is direction-specific under replacement, and is maintained by distributed aligned writes.

We ran the geometric comparison against these objects on Phi-2. Mid-stack, $\cos(b_L, \cdot)$ against the first-token direction at its emergence layer (located here at layer 21, peak coordinate magnitude ~ 1250 ; cf. Shi et al., 2026, who propose the first-token massive activation as a “shared global reference vector” for attention) is 0.04–0.17, and against the residual-sink and massive-coordinate directions ≤ 0.27 : the objects are distinct. The separation is trained: in the random-init twin the reference *is* the sink blob ($\cos = 0.98$, no massive activation), and training rotates the reference off the sink. On outlier dimensions: b_L carries 10–23% of its squared norm on the twenty largest last-layer outlier coordinates against an isotropic share of 0.8%, a real but minority overlap that grows toward the readout; its own dominant coordinates are largely disjoint from the outlier set (top-20 Jaccard 0.05–0.14). Mid-stack, b_L is a distributed direction with a minority frequency-correlated component, not the axis-aligned outlier set, and it should not be conflated with the sink objects of Qiu et al. (2026) or Shi et al. (2026).

Entropy and confidence regulation. Stolfo et al. (2024) identify neurons that regulate output confidence partly through the unembedding null space. The entropy-minimum observation of §7 is adjacent in effect but attaches to a carried stream component rather than to specific neurons; we made no attempt to connect the two.

Bias as coupling. Treating bias as a weight onto a constant input is the classical homogeneous-coordinates construction. Weight-based taxonomies of gated neurons (Gerstner and Schütze, 2025) and readings of MLP units as binary routers (Balogh, 2026) characterise gate function from the parameters, and massive activations have been read as carried constants with a bias-like role (Sun et al., 2024, 2026). The observation here joins these: a trained transformer implements the homogeneous-coordinates construction spontaneously, with the constant supplied by the network’s own maintained mean rather than by a dedicated input. Sun et al. (2024) also report the trained converse on the attention side: augmenting self-attention with explicit learnable bias parameters yields models that do not form massive activations: given the parameter, the network uses it instead of building a carried constant. OPT plays the same role on the gate side here (§5): the parameter-based strategy exists, and most families do not take it. Finally, modern architectures dropped dense biases for training stability with no measured quality loss (Chowdhery

et al., 2022); the decomposition here offers an account of why the removal is functionally free on the gate side, namely that the operating point never lived in b_n .

10 Limitations

- **Replication scope.** The static decomposition (§5) is shown on eight further models and is absent in OPT, where an opposing LayerNorm bias cancels the carried reference; the ReLU-versus-OPT-family confound (every sampled ReLU model is an OPT) is not separated by matched-nonlinearity twins. The causal dose-response and the entropy release arm replicate on two further families (Pythia-1.4B, Qwen2.5-1.5B); the replacement panel (§4, one site, L10) and co-removal (§6) are Phi-2-only. The formation results, including the checkpoint time-course, are on one model (Pythia-410M), one seed, trend-grade (15–30 documents).
- **Single-direction geometry is family-dependent.** The cross-layer “nearly a single direction” claim (§2) is heterogeneous across the replication families: $\cos(\hat{m})$ between mid-stack layers is high in GPT-2-medium (0.95) and Qwen2.5-1.5B (0.96), modest in TinyLlama-1.1B (0.74), and below its own twin floor in Pythia-1.4B (0.41 vs. twin 0.67). The coupling mechanism replicates in every family; the single-carried-direction geometry does not, and is not claimed to.
- **Within-layer atomicity is open.** b_L is a corpus mean, rank-1 by construction; a single mean vector would look identical whether one atomic direction or several register-conditioned bias lines were carried underneath it. Preliminary evidence (structured, register-separable secondary components surviving removal of the $\hat{m}$ projection) points toward the latter; whether those secondary lines carry any independent gating function is left to separate work, and every causal result here concerns only the rank-1 mean.
- **The reference is register-adaptive, not frozen.** Across prose and code (four replication families; Phi-2 itself was not tested across registers) the coupling mechanism transfers: $\rho(\text{resting, duty})$ on code matches prose layer by layer (0.75–0.99). But the direction shifts partially with register: in the RMSNorm models (where the twin cosine floor is low enough to discriminate), trained $\cos(b_L^{\text{prose}}, b_L^{\text{code}}) \approx 0.8$, code $|b_L|$ runs up to $1.4\times$ prose, and cross-register rank transfer (≈ 0.75 –0.85) sits below the within-register ~ 0.95 . “Carries and maintains a constant” should be read as a register-adaptive reference, not a corpus-independent vector.
- **Operating-point asymmetry.** The pre-registered monotone knob failed (§7); the entropy minimum that replaces it rests mainly on the release direction, since the amplification arm is shallow in Phi-2 and varies across families.
- **Manipulation scope.** The reference is the corpus mean; a position-resolved background is a refinement not tested. Manipulations leave positions < 16 untouched, so attention re-import from the sink region is possible. α -scaling changes the LayerNorm input distribution, so all downstream effects pass through LN renormalisation; the norm-matched control shares this property, and the norm-preserving replacement panel (§4) excludes a pure gain-boost account at L10, the one site where it was run. The function-word replacement is one naive construction of a token background, not an optimised one.
- **Geometric comparisons are activation-proxy.** The comparisons against first-token massive activations, the residual sink, and last-layer outlier dimensions (Related work) use activation-

based proxies of each object on one model; a weights-only unification against each paper’s exact construction is not attempted.

11 Reproducibility

Scripts and JSON artifacts (public checkpoints, fixed seeds):

`census_background_reference.py` (§2, §3), `census_background_identity.py` (§4, co-removal), `census_reference_mechanism.py` (§5, §6, §7), `census_checkpoint_trajectory.py` and `census_formation_microscopy.py` (§8), `census_reference_geometry.py` (geometric comparisons, Related work), `census_crossmodel_causal.py` (cross-model dose-response and entropy V), `census_bias_migration_scale.py` (Table 4), `census_bias_migration_formation.py` (coupling time-course over checkpoints), `census_bias_migration_nonlinearity.py` (eight-model scan, OPT boundary), `census_reference_norm_decomp.py` (residual-versus-LN-bias decomposition), `census_suppressor_reference.py` (read-side alignment), `census_reference_corpus.py` (register adaptivity), `census_readweight_factorization.py` (per-gate anatomy). Each experiment’s pre-registration, gates, and full numbers are in the project’s lab notebook (`plan_mid_stack_empirical_basis.md`, §3t–§3x). Code and data are available at <https://github.com/EvidentSolutions/llm-interp/tree/main/background>.

Use of AI Assistants

Large language models were used as assistive tools for coding, running experiments, and drafting text. All research questions, experimental design, and reported claims were directed and verified by the authors, who take full responsibility for the content.

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